

1 Material and methods

Neo gia — muc that nam o ban chinh.

1.1 No-anchor arm

Neo gia (E1).

1.2 Expected-value probes

Neo gia (E2).

2 Best-response profiling against known opponents

Every result so far comes from one model playing copies of itself, a design that cannot separate a model reasoning about its opponents from a model following a script. We remove the symmetry: a single LLM is seated against five opponents whose policies we fix in advance. Because the opponents are then known, the best response can be computed exactly, and “does the model best-respond?” becomes arithmetic.

2.1 Design

Seat 1 is an LLM; the other 5 run one deterministic policy, identical across seats and constant within a game — ALL-DEFECT contributes 0 every round, ALL-CONTRIBUTE 4, FAIR-SHARE 2 (the per-round equal split of the target, which the prompt names), and COND. COOP. the round-average of the other five seats, rounded to a legal amount. Crossing these with 5 risks $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, 5 models and 10 repetitions gives 100 cells and 1000 games; the game is otherwise that of Section 1. Only one seat calls a model, so a game costs 10 API calls rather than 60.

Two protocol facts govern every reading below. The LLM is *not* told that its opponents are deterministic, scripted or identical; it receives the prompt of the symmetric game, in which the pool is never printed but the complete per-seat history is shown each round in fixed positions, so the pool is recoverable by addition. That history is what keeps the omission from being decisive: under ALL-DEFECT and ALL-CONTRIBUTE the five scripted seats return the same five numbers every round, so a player entering round 6 has seen 5 identical rounds and the opponents’ policy is an induction from the transcript rather than a fact we withheld — and one the models demonstrably perform (Figure 1). We therefore separate round 1, where no history exists and nothing can be inferred, from rounds 6–10, and rest the claim on the latter. And before computing anything we replayed all five scripted seats of all 1000 games from the LLM’s recorded moves, requiring a round-by-round match; all replay exactly, which is what licenses treating the opponents as known.

2.2 What a rational seat would do

With deterministic opponents the LLM faces a decision problem, solved by exhaustive search over all 3^{10} trajectories under the game’s own scoring rule.

Proposition 1. *Against ALL-DEFECT and ALL-CONTRIBUTE the best response is to contribute 0 at every p . Against FAIR-SHARE and COND. COOP. it is $0 \rightarrow 20$, switching at $p^* = 0.5$, where both are optimal.*

Proof sketch. The scripted seats bank 0, 200 and 100 in the first three profiles. Against ALL-DEFECT the group total cannot exceed the LLM’s own 40, so 120 is unreachable and every unit contributed is lost; against ALL-CONTRIBUTE the target is secured before the LLM moves, so again every unit is lost. Against FAIR-SHARE the group needs exactly 20 from the LLM, buying a certain 20 against a lottery worth $(1 - p)40$, equal at p^* . ■

The fourth profile is not an independent condition: a conditional cooperator averages four conditional cooperators sitting at 2 and the LLM playing $c \in \{0, 2, 4\}$, giving $(8 + c)/5 \in \{1.6, 2.0, 2.4\}$, all of which round to 2, and enumerating all 59,049 LLM trajectories confirms none ever moves such a seat off 2. Its agreement with FAIR-SHARE across all 5 models is a consistency check, not a replication.

2.3 Results

No model best-responds where the answer is forced. We state the result on the 500 games in which contributing is *strictly dominated* — ALL-DEFECT, where the target cannot be reached, and ALL-CONTRIBUTE, where it is already secured — because there keeping a unit beats spending it under *every* increasing utility, so no assumption about risk attitude is doing any work, and because the opponents’ policy is legible in the transcript. Not one of those 500 games, and not one of the 1000 overall, has the LLM seat contribute nothing across the ten rounds (0 and 0 respectively); the smallest total observed is 4 under ALL-DEFECT and 2 under ALL-CONTRIBUTE, and the mean expected payoff forfeited is 12.78 of the 40-unit endowment (Table 1). Splitting by round does not rescue it. In round 1, with no history to read, the seat pays 2.21 units, which we do not count against it; over rounds 6–10, after 5 rounds in which the same five opponents made the same five moves, it still pays 6.57 units per game, worth 4.82 in expectation, and only 38% of games end clean (Figure 1). The spread across models is wide enough to be the finding: over that stretch Luna pays 0.32 units and is clean in 92% of games, whereas Qwen pays 13.12 and Grok 16.00, neither with a single clean game. Under ALL-DEFECT the target is never met, so all six seats face the same lottery and the forfeited column is exactly the margin by which a seat that does nothing out-earns the LLM beside it, from 3.72 for Luna to 15.05 for Qwen. A trivial free-rider profits from every model in the panel.

Table 1: Distance to best response. *Gap* is the distance from the realised total contribution to the nearest best response, in contribution units; *Forfeited* prices that gap as expected payoff under the best response minus expected payoff under what was played, in units of the 40-unit endowment, hence a risk-neutral valuation; *Optimal* is the percentage of games played at zero cost. Expectations are taken over the catastrophe lottery, so the numbers do not inherit its variance. [†]These two blocks hold fewer independent numbers than columns and should be read as a single measurement: the best response there is 0 at every p , so *Gap* is by construction the mean total contribution, and under ALL-CONTRIBUTE *Forfeited* equals *Gap* to the cent, the target being secured whatever the seat does. Only the FAIR-SHARE and COND. COOP. blocks carry two independent numbers.

Opponents	Model	Gap	Forfeited	Optimal
All-defect [†]	Haiku 4.5	7.76	3.73	0%
	Flash-Lite 3.5	13.88	6.52	0%
	GPT-5.6 Luna	8.48	3.72	0%
	Qwen3-235B	29.88	15.05	0%
	Grok-4.20 NR	30.68	14.71	0%
All-contribute [†]	Haiku 4.5	15.32	15.32	0%
	Flash-Lite 3.5	10.92	10.92	0%
	GPT-5.6 Luna	4.52	4.52	0%
	Qwen3-235B	20.40	20.40	0%
	Grok-4.20 NR	32.92	32.92	0%
Fair-share	Haiku 4.5	9.28	6.36	34%
	Flash-Lite 3.5	8.00	4.80	60%
	GPT-5.6 Luna	8.12	5.15	54%
	Qwen3-235B	8.28	13.26	6%
	Grok-4.20 NR	17.92	14.72	32%
Cond. coop.	Haiku 4.5	9.28	6.40	36%
	Flash-Lite 3.5	8.00	4.80	60%
	GPT-5.6 Luna	8.12	5.29	56%
	Qwen3-235B	8.28	13.26	6%
	Grok-4.20 NR	18.00	14.80	28%

The ALL-CONTRIBUTE trap separates cooperation from compliance. Here the scripted seats bank 200 alone, so the target is secured whatever the LLM does and every unit it contributes is destroyed with certainty: contributions are not cooperation, because there is nothing left to cooperate towards. All five models pay, from 4.52 of the 40-unit endowment for Luna to 32.92 for Grok. This profile also supports a measurement immune to the objection that we score models with knowledge we withheld. In 224 of the 250 games the pool shown in the prompt’s own history reaches 120 by the end of round 6, and earlier in the rest; from the next round on, no belief about the opponents can justify a contribution. Counting only what is paid after that point, the waste is 5.09 per game, or 30.3% of everything contributed here — and it is concentrated. Luna and Flash-Lite pay nothing at all after the outcome is settled, in 100% and 100% of games; Haiku wastes 1.12; Qwen wastes 8.08 and Grok 16.24, neither with a single clean game. For Grok that

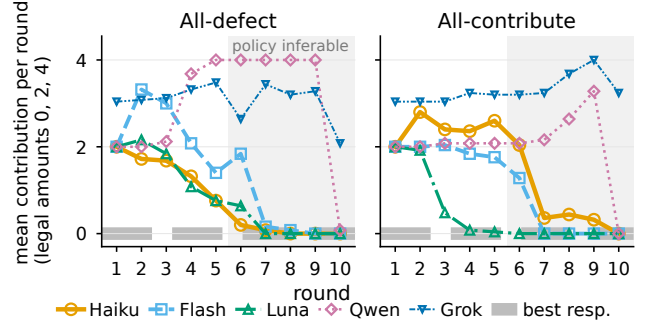


Figure 1: Mean contribution in each round, on the two profiles where contributing is strictly dominated, so that the best response is 0 in every round; $n = 50$ games per line. Shading marks rounds 6–10, by which the transcript has repeated 5 identical rounds of opponent moves and the opponents’ policy is an induction rather than a fact we withheld; over that stretch the panel still pays 6.57 units per game and only 38% of games end clean. The shape is what the tables cannot show: some models descend to the best response and stay there, whereas Qwen climbs as the evidence accumulates and Grok never adjusts downwards. Nested line widths and unfilled markers keep models playing identical values visible — 4 of 5 open at exactly 2.00 — without displacing any series.

is 49.3% of its total contribution, spent after the game was provably over.

Models anchor and adjust; they do not compute. Four of five models open at exactly the equal split of 2.00 in 100% of games (Haiku, Flash, Luna, Qwen), Grok averaging 3.03. Deleting the two prompt spans that state that split (§1.1) does not return a single verdict: Qwen and Grok move sharply, Qwen from 2.00 to 3.40 and Grok from 3.17 to 4.00 ($P < 0.001$ and $P < 0.001$, permutation tests clustered by game), whereas Haiku, Flash and Luna do not move at all (largest shift 0.04 of a unit, smallest of their p -values 0.111). The split is a focal point two models take from the wording and three reconstruct without it, as they can, since the target, group size and round count remain and fix it by division. What follows the opening is adjustment, not optimisation. Against ALL-CONTRIBUTE, where nothing should be contributed from round 1, Luna falls 2.00 $\rightarrow$ 1.92 $\rightarrow$ 0.48 $\rightarrow$ 0.08 and is below one unit by round 3, while Haiku and Flash hold out to round 7. The two that never adjust make the point from the other side: across rounds 7 to 9 Qwen rises from 2.16 to 3.28 and Grok from 3.24 to 4.00, *increasing* payment over exactly the rounds in which the visible record already showed the target met.

A fixed end-game rule becomes a population-level failure. Qwen plays 0 in the final round in 100% of ALL-CONTRIBUTE, 98% of ALL-DEFECT, 100% of FAIR-SHARE and 100% of COND. COOP. games: of 5 models it is the only one whose end-game rule is uncon-

ditional. The others condition on the state, withdrawing when it is free (Haiku, Flash and Luna all at 100% under ALL-DEFECT and ALL-CONTRIBUTE) but paying on when the target depends on it (16%, 0% and 2%). The cost surfaces only in the population that exposes it: under FAIR-SHARE the group needs exactly 20 from the LLM seat, and Qwen contributes 2 for nine rounds then nothing in 47 of 50 games, leaving the group exactly 2 units short in each — 118.1 against 120 on average — and reaching the target in 6% of games against 100% for Flash-Lite. A heuristic that costs nothing in two populations destroys the public good in a third, and self-play would not have revealed it.

Policies are degenerate, and risk barely moves them. Under FAIR-SHARE Flash-Lite plays one identical trajectory in all 50 of its 50 games, while Grok is bimodal, contributing 2 every round in 24 games and 4 every round in 24. Fitting a slope in p per cell and testing it by permutation, 5 of 20 cells reach the 5% level, all of them in the two profiles where the best response is 0 at every p and risk is irrelevant; three have the wrong sign, Grok paying 7.9 more units per unit of p against all-defectors ($P = 0.002$), Luna 6.5 and Flash-Lite 5.2, while Qwen alone moves the other way (-1.4). Contributions are flat under FAIR-SHARE and COND. COOP., which is where flatness is the failure, because there the optimum does switch at p^* (Figure 2).

2.4 What this measures, and what it does not

Pricing a gap in expected payoff assumes risk neutrality, and under FAIR-SHARE and COND. COOP. that assumption carries the result rather than describing it: there the choice is a certain 20 against a lottery worth $(1-p)40$, and 41% of the pooled forfeiture comes from those two profiles alone. Under constant relative risk aversion $u(x) = x^{1-\gamma}/(1-\gamma)$ the switch point moves from $p^* = 0.50$ at $\gamma = 0.0$ to 0.29 at $\gamma = 0.5$ and 0.07 at $\gamma = 0.9$, and identical play is scored a best response in 186, 247 and 308 of 1000 games respectively — the pooled figure is in that range a statement about γ , not about the models. We therefore declare the coefficient we assume, zero, treat the pooled forfeiture of 10.83 as an upper bound, and rest the claim on the 500 dominated games, where the count of best responses stays at 0 for every γ we tried. Likewise the count of exactly optimal games, 186 of 1000, is a property of a risk-neutral scorer; what survives any scorer is that every one of them totalled exactly 20, so the models land on the optimum only where it coincides with the focal point. The gap is also computed with information the model was never given, since it is told neither that its opponents are scripts nor which; that is why the claim rests on rounds 6–10, where the transcript has by then repeated the opponents’ policy 5 times, and on the post-proof measure, which uses only what the prompt displays. Where neither device applies — round 1, and the FAIR-SHARE and COND. COOP. profiles, whose opponents must be extrapolated rather than merely read

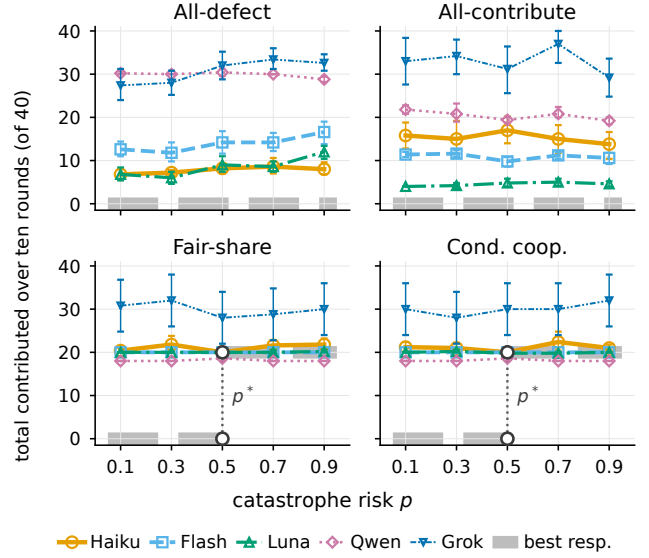


Figure 2: Total contributed by the LLM seat over the ten rounds, of the 40-unit endowment, against catastrophe risk; one panel per opponent profile, $n = 10$ games per point, with percentile bootstrap 95% intervals. The thick grey dashed line is the best response of Proposition 1, drawn beneath the model lines so that models sitting exactly on it stay visible; under FAIR-SHARE and COND. COOP. it steps from 0 to 20 at $p^* = 0.5$, and the two open circles mark the tied pair there, either of which scores a zero gap. Flash-Lite’s intervals vanish under FAIR-SHARE because all 50 of its games are identical — no variation to express, rather than none existing. Every marker sits at its true risk level; where models play the same value, nested line widths and unfilled markers keep both visible.

— we report the gap but do not lean on it. The shortfall is not mere misreading: the panel answers in-situ rule probes at 100.0% and the expected-value comparison at 72.4% (§1.2). Finally, each cell holds 10 games, so every interval is a percentile bootstrap and every p -value a permutation test with 10,000 resamples under a fixed seed, clustered on the game; claims about one model in one profile rest on 50. The game-level randomisation is seeded and reproduces, but the model text does not — replaying one model’s cells at identical seed and temperature returned different play in 31 of 54 cells (57%) — so we release prompts, configurations, seeds and code, and promise reproducibility of the *design* rather than of the transcripts. Two of our findings are nulls, and they are the section’s most secure results rather than failures to find an effect: that no model best-responds where every alternative is strictly dominated is a positive claim about 500 games with a known right answer, and ALL-CONTRIBUTE is built so that its null admits one reading.